Human Factors: Technical Worksheet

Grade 12 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Human factors studies how humans perform with tools and systems—perception, cognition, error modes, and environment—so designs prevent harm and support reliable performance.

Do this:

1. **Give an example** where a confusing interface contributed to a mistake.

   _____

2. **Circle** a human-factors aim: Design for predictable human performance · Assume perfect users

Part 2 – Error types and defenses (about 5 minutes)

Slips (right plan, wrong execution), mistakes (wrong plan), and lapses (memory failures) need different defenses: constraints, checklists, confirmations, training.

> **Real-life scenario:** A student intends to unshare one file (plan OK) but clicks “Anyone with the link” for the whole folder (slip). A confirm dialog naming the scope would help.

Discuss with a partner or reflect in writing: Is that failure best fixed by “try harder” or by better constraints?

Do this: Circle the better pairing.

Defense	Best for
Checklist before deploy	Mistake-prone planning · Irrelevant
Confirm dialog with consequences	Slip on destructive action · Never useful
Remove almost-identical icons	Perception slip · Purely cosmetic always

Part 3 – Mental models and feedback (about 5 minutes)

Users act on **mental models**. If the system’s feedback does not match the model, performance collapses. Visibility of system state is a human-factors essential.

> **Real-life scenario:** A cloud sync icon looks “done” while upload still runs. The laptop closes; a draft never reaches the teacher.

Critical thinking: What state signals should the UI show?

Do this:

1. **Describe a misleading status indicator** you have seen.

   ---

2. **Propose a clearer signal** (text + icon behavior).

Part 4 – Workload, stress, and performance (about 5 minutes)

Under high workload or stress, tunnel vision and skill regression appear. Critical systems should simplify choices, shed nonessential alarms, and support recovery.

> **Real-life scenario:** During a timed outage drill, five red alerts flash at once with similar wording. Operators freeze.

Discuss with a partner or reflect in writing: How would you prioritize alerts using human-factors thinking?

Do this: Circle **Yes** or **No**.

Design under stress	Supportive?
Tier alerts by severity with distinct cues	Yes · No
Flood identical alarms	Yes · No
Provide a clear recommended first action	Yes · No

Part 5 – Human-factors case card (about 5 minutes)

Do this: Build a case card: Context → Human limit/factor → Failure → Design defense. Use a school or public example.

State one measurable improvement you would expect.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

□ I can define **human factors** in systems.

□ I distinguish slips, mistakes, and lapses at a basic level.

□ I value feedback that matches mental models.

□ I design simpler paths under stress/workload.

□ I can propose a concrete design defense.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Human Factors (Technical). Central topic: Human factors. Main technology area: Systems design and safety.